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Paper 1: Translational Modeling of Gut Microbiome-Mediated Drug Metabolism: A Case Example

Bibliographic Information

- **Title:** Translational Modeling of Gut Microbiome-Mediated Drug Metabolism: A Case Example of Sulfasalazine
- **Authors:** Cheng L, Cai J, et al.
- **Journal:** CPT: Pharmacometrics & Systems Pharmacology, 2026, 15(4), e70246
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Quick Take

This paper presents a novel PBPK modeling framework to predict gut microbiome-mediated drug metabolism using sulfasalazine as a case study. It combines in vitro fecal incubation kinetics with bacterial count scaling and Simcyp PBPK models to predict metabolite PK. The approach shows good predictive performance (weighted average fold-errors ~ 1.2 for C_{max} , T_{max} , AUC) and offers a template for other drugs undergoing gut microbial metabolism.

Executive Summary

The study develops a translational modeling framework integrating in vitro gut microbiome metabolism kinetics with PBPK modeling to predict in vivo pharmacokinetics of sulfasalazine and its metabolites. Using pooled human fecal homogenate incubations, Michaelis-Menten parameters were determined and scaled to colonic metabolism using a colon-to-feces bacterial count ratio of 0.47. PBPK models for sulfasalazine, sulfapyridine, and mesalamine were built and verified against clinical data. The linked model predicted metabolite PK with weighted average fold-errors of 1.21, 1.22, and 1.05 for C_{max} , T_{max} , and AUC , respectively. Sensitivity analyses highlighted the importance of colon permeability and transit time. This framework enables prospective prediction of gut microbiome-mediated drug metabolism and can guide prodrug design and dose prediction.

Scientific Context & Motivation

The gut microbiome is recognized as a contributor to drug metabolism, yet quantitative translation from in vitro to in vivo is lacking. Sulfasalazine is a classic example of a prodrug activated by bacterial azoreductases in the colon, but no established IVIVE framework existed for gut microbial metabolism. The challenge lies in deconvoluting the complex interplay among microbial metabolism, GI transit, and drug absorption. This paper fills the gap by using a mechanistic PBPK approach (Simcyp ADAM model) with bacterial count scaling to predict metabolite formation and absorption, providing a template for other drugs.

✂ Methodological Snapshot

The study uses a middle-out PBPK modeling approach in Simcyp V22 with the ADAM absorption model. In vitro Michaelis-Menten kinetics for sulfasalazine metabolism were measured in pooled human fecal homogenate. V_{max} was scaled to colonic metabolism using a colon-to-feces bacterial count ratio of 0.47 derived from matched samples in healthy subjects. Separate PBPK models for sulfasalazine, sulfapyridine, and mesalamine were built and verified against clinical PK data when each compound was dosed alone. The models were then linked by activating colon luminal metabolism in the sulfasalazine model to drive metabolite formation. Sensitivity analyses explored the impact of luminal V_{max} , colon P_{eff} , and colon transit time on metabolite PK.

☒ Structural Model Breakdown

The sulfasalazine PBPK model includes: ADAM absorption with regional P_{eff} (jejunum 0.32, colon 0.032×10^{-4} cm/s), full PBPK distribution (V_{ss} 0.09 L/kg), hepatic metabolism (CL_{int} 18.145 μ L/min/mg), biliary clearance (genotype-dependent), BCRP efflux in gut (J_{max} genotype-dependent), and colon luminal metabolism (Michaelis-Menten with V_{max} 7061 pmol/min/mg luminal content, K_m 1000 μ M). Sulfapyridine model: ADAM absorption (jejunum P_{eff} 2.34, colon 1.17×10^{-4} cm/s), full PBPK distribution (V_{ss} 0.71 L/kg, tissue K_p scalar 1.65), NAT2 metabolism (CL_{int} genotype-dependent: 0.20–1.13 μ L/min/mg cytosolic protein), renal clearance 0.50 L/h. Mesalamine model: ADAM absorption (jejunum P_{eff} 3.3, colon 0.132×10^{-4} cm/s), full PBPK distribution (V_{ss} 0.23 L/kg), hepatic clearance 23 L/h. The linked model connects sulfasalazine colon metabolism to sulfapyridine and mesalamine formation in the colon lumen, with subsequent absorption via their respective P_{eff} values.

Detailed Methodological Analysis

Modeling Approach

Physiologically-based pharmacokinetic (PBPK) modeling using the Simcyp Simulator V22 with the Advanced Dissolution, Absorption and Metabolism (ADAM) model. Full PBPK distribution models were used for all compounds. The Segregated Transit Time (STT) model was used for colon transit (ascending colon MRT). Luminal metabolism was implemented as a first-order or Michaelis-Menten process in the colon only.

Data Sources

In vitro: pooled fresh fecal homogenate from 4 healthy donors incubated with sulfasalazine (5–5000 μ M) for 20 min. Bacterial counts: colon luminal samples and feces from 9 healthy subjects collected via a non-invasive device. Clinical PK data: digitized from published studies (10+ studies) covering doses 250–4000 mg, various BCRP and NAT2 genotypes, and different formulations (immediate release, Eudragit L/S coated).

Estimation Methods

In vitro kinetic parameters (V_{max} , K_m) estimated by nonlinear regression of formation rates vs. concentration. PBPK model parameters were either predicted (e.g., P_{eff} from gMDCK perme-

ability), taken from literature, or optimized manually to match clinical data (e.g., BCRP J_{max} , NAT2 CL_{int}). No formal population PK estimation was used; simulations were deterministic with inter-individual variability from Simcyp default populations.

Model Evaluation

Model performance evaluated by comparing simulated vs. observed concentration-time profiles and PK parameters (C_{max} , T_{max} , AUC). Fold-error (observed/predicted) calculated for each study; weighted average fold-errors reported. Visual predictive checks (5th, 50th, 95th percentiles) shown for representative studies. Sensitivity analyses conducted for key parameters.

Covariate Analysis

Covariates included BCRP genotype (ABCG2 421C/C, C/A, A/A) affecting J_{max} and biliary clearance, and NAT2 genotype (slow, intermediate, fast acetylator) affecting cytosolic CL_{int} . These were incorporated as fixed effects in the PBPK models. No formal covariate selection was performed; genotypes were assigned based on reported study populations.

Statistical Rigor Assessment

The study uses a deterministic PBPK approach with inter-individual variability from Simcyp default populations (CV for demographics, enzyme abundances). No formal statistical inference (e.g., confidence intervals) is provided for parameter estimates. Model evaluation relies on fold-error criteria (< 2 -fold considered acceptable) and visual inspection of concentration-time profiles. Sensitivity analyses are thorough but qualitative. The sample size for bacterial count scaling ($n=9$) is modest but adequate for a ratio estimate. Missing data (e.g., genotype not reported in some studies) were handled by using default wildtype values. Overall, the statistical rigor is appropriate for a proof-of-concept translational modeling study, but formal uncertainty quantification would strengthen future applications.

Key Findings

In vitro Michaelis-Menten parameters: sulfapyridine formation $V_{max} = 650.5$ pmol/min/mg feces, $K_m = 3648$ μ M; mesalamine formation $V_{max} = 200.9$ pmol/min/mg feces, $K_m = 1605$ μ M. Colon-to-feces bacterial count ratio geometric mean = 0.47 (%CV 56%). PBPK models predicted parent and metabolite PK within twofold error across multiple clinical studies (doses 250–4000 mg, various BCRP and NAT2 genotypes). Weighted average fold-errors for sulfapyridine and mesalamine after sulfasalazine dosing: C_{max} 1.21, T_{max} 1.22, AUC 1.05. Sensitivity analysis showed metabolite PK is sensitive to luminal V_{max} , colon P_{eff} , and colon transit time. At therapeutic doses (2000–4000 mg), conversion is high but not complete; saturation kinetics important at high doses. Colon absorption of sulfasalazine is negligible due to low permeability and rapid metabolism.

🔗 Clinical & Regulatory Implications

The framework enables prediction of metabolite exposure for prodrugs activated by gut microbiome, guiding dose selection and formulation design. For sulfasalazine specifically, the model confirms that colon metabolism is the primary source of active metabolites, and that BCRP and NAT2 genotypes significantly affect parent and metabolite PK. The sensitivity analysis suggests that at therapeutic doses, conversion is high but not complete, so changes in microbiome activity (e.g., due to antibiotics) could alter metabolite exposure. The approach can be extended to other drugs undergoing gut microbial metabolism (e.g., deglucuronidation of UGT substrates) and can inform regulatory submissions for prodrugs. The colon P_{eff} estimation method using dog data provides a practical solution for a parameter that is difficult to measure in humans.

Strengths & Limitations

Strengths

- First study to combine IVIVE of gut microbiome metabolism kinetics with PBPK modeling for prospective prediction of parent and metabolite PK
- Extensive verification against multiple clinical studies with different doses, genotypes, and formulations
- Use of matched colon and fecal bacterial counts from healthy subjects for scaling, reducing variability
- Comprehensive sensitivity analyses exploring impact of luminal V_{max} , colon P_{eff} , and transit time
- Practical framework readily applicable to other drugs undergoing gut microbial metabolism (e.g., deglucuronidation, prodrug activation)

Limitations (Acknowledged by Authors)

- K_m value capped at 1000 μM in Simcyp, lower than measured in vitro; saturation effects approximated by scaling V_{max} to maintain V_{max}/K_m ratio
- Azoreductase expression assumed proportional to total bacterial count; actual enzyme abundance not measured
- Colon effective permeability (P_{eff}) derived from dog intraluminal dosing studies, not directly measured in humans
- Inter-individual variability in microbiome composition not fully captured; only %CV of bacterial count ratio used
- Limited mesalamine PK data after sulfasalazine dosing (only two studies)

Limitations (Expert Review)

- Only one drug example (sulfasalazine) tested; generalizability to other microbial reactions (e.g., reduction, hydrolysis) needs validation
- No inclusion of diseased populations (e.g., ulcerative colitis patients) where microbiome composition may differ
- Assumption of uniform bacterial activity throughout colon; regional differences in azoreductase expression not considered

- In vitro incubation conditions (aerobic shaker, 20 min) may not fully replicate anaerobic colonic environment
- No formal model selection or uncertainty quantification for PBPK parameters

Generalizability

The framework is designed to be generalizable to other drugs metabolized by gut microbiome, provided in vitro kinetic data and bacterial count scaling are available. The approach is particularly suited for prodrugs activated in the colon and for drugs undergoing deglucuronidation. However, validation with additional compounds is needed to confirm the scaling factors (e.g., bacterial count ratio, colon P_{eff} estimation method).

Key Equations

Michaelis-Menten kinetics for metabolite formation

$$v = \frac{V_{max} \cdot [S]}{K_m + [S]}$$

Describes the rate of sulfapyridine or mesalamine formation from sulfasalazine by bacterial azoreductases in fecal homogenate incubations.

Scaling of Vmax from feces to colon

$$V_{max,colon} = V_{max,fecal} \times \frac{\text{bacterial count}_{colon}}{\text{bacterial count}_{feces}}$$

Scales in vitro V_{max} measured in pooled fecal homogenate to in vivo colonic metabolism using the ratio of geometric mean bacterial counts (0.47).

Figures & Tables

- **Figure 1:** Metabolism pathway of sulfasalazine and schematic of the PBPK modeling workflow.
 - *Significance:* Illustrates the biotransformation of sulfasalazine to sulfapyridine and mesalamine by bacterial azoreductases, and the stepwise model development and verification process.
- **Figure 2:** Representative simulated vs. observed concentration-time profiles for sulfasalazine (A), sulfapyridine in slow (B) and fast (C) acetylators, and mesalamine immediate release (D) and modified release (E,F) formulations.
 - *Significance:* Demonstrates the verification of individual PBPK models for each compound when dosed alone, showing good predictive performance within 5th-95th percentiles.
- **Figure 3:** Simulated vs. observed concentration-time profiles of sulfasalazine, sulfapyridine, and mesalamine after oral dosing of sulfasalazine in healthy subjects.
 - *Significance:* Shows the performance of the linked PBPK model with colon luminal metabolism, capturing both parent and metabolite PK across different studies.
- **Figure 4:** Sensitivity analysis heatmaps and plots showing impact of luminal V_{max} , colon P_{eff} , and colon transit time on sulfapyridine and mesalamine PK parameters.

- *Significance*: Identifies key drivers of metabolite exposure: formation rate, colon permeability, and transit time; explains why mesalamine has lower systemic exposure than sulfapyridine.
 - **Table 1**: PBPK input parameters for sulfasalazine, sulfapyridine, and mesalamine models.
 - *Significance*: Provides all model parameters including physicochemical properties, absorption, distribution, elimination, and transporter kinetics; essential for reproducibility.
 - **Table 2**: Verification of individual PBPK models: observed vs. predicted C_{max} , T_{max} , AUC for sulfasalazine, sulfapyridine, and mesalamine when dosed alone.
 - *Significance*: Demonstrates that each compound model is adequately verified before linking, with fold-errors generally within twofold.
 - **Table 3**: Observed vs. predicted PK parameters for sulfasalazine, sulfapyridine, and mesalamine after oral dosing of sulfasalazine across multiple clinical studies.
 - *Significance*: Main results table showing the linked model performance across different doses, genotypes, and populations; weighted average fold-errors reported.
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Supplementary Materials

Supplementary materials include LC-MS method details, additional PK verification plots (Figures S1–S7), sensitivity analysis results (Figure S8), and bacterial count data (Table S2). These support the main findings and provide further validation.

Future Directions

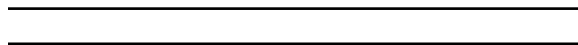
Future work should validate the framework with additional drugs undergoing different microbial reactions (e.g., deglucuronidation, reduction). Direct measurement of azoreductase abundance in colon vs. feces could refine scaling. Inclusion of diseased populations (e.g., IBD) would assess impact of altered microbiome. Extension to other PBPK platforms and incorporation of microbiome dynamics (e.g., antibiotic effects) would increase utility. Prospective application in drug discovery for prodrugs is a natural next step.

Expert Commentary

This study represents a significant step forward in quantitative microbiome pharmacology. The use of matched colon and fecal bacterial counts is a clever way to address the long-standing challenge of scaling in vitro fecal data to in vivo colonic metabolism. The sensitivity analyses are particularly insightful, showing that metabolite PK is a net result of formation rate, colon permeability, and transit time. The framework is pragmatic and leverages existing Simcyp capabilities, making it accessible to many pharmacometricians. A key teaching point is the importance of obtaining full Michaelis-Menten kinetics rather than assuming linear clearance, especially at high luminal concentrations. The assumption that azoreductase activity scales with total bacterial count is reasonable but should be tested with enzyme-specific data. Overall, this work sets a new standard for modeling gut microbiome-mediated drug metabolism.

Bottom Line

This paper provides a robust translational PBPK framework for predicting gut microbiome-mediated drug metabolism, using sulfasalazine as a proof-of-concept. The approach combines in vitro fecal incubation kinetics, bacterial count scaling, and mechanistic absorption modeling to predict metabolite PK with good accuracy. It is ready for application to other drugs and can guide prodrug design, dose prediction, and assessment of microbiome-related variability. Pharmacometricians should adopt this framework for compounds where gut microbial metabolism is suspected, and further validate with additional case studies.



☒ Figures

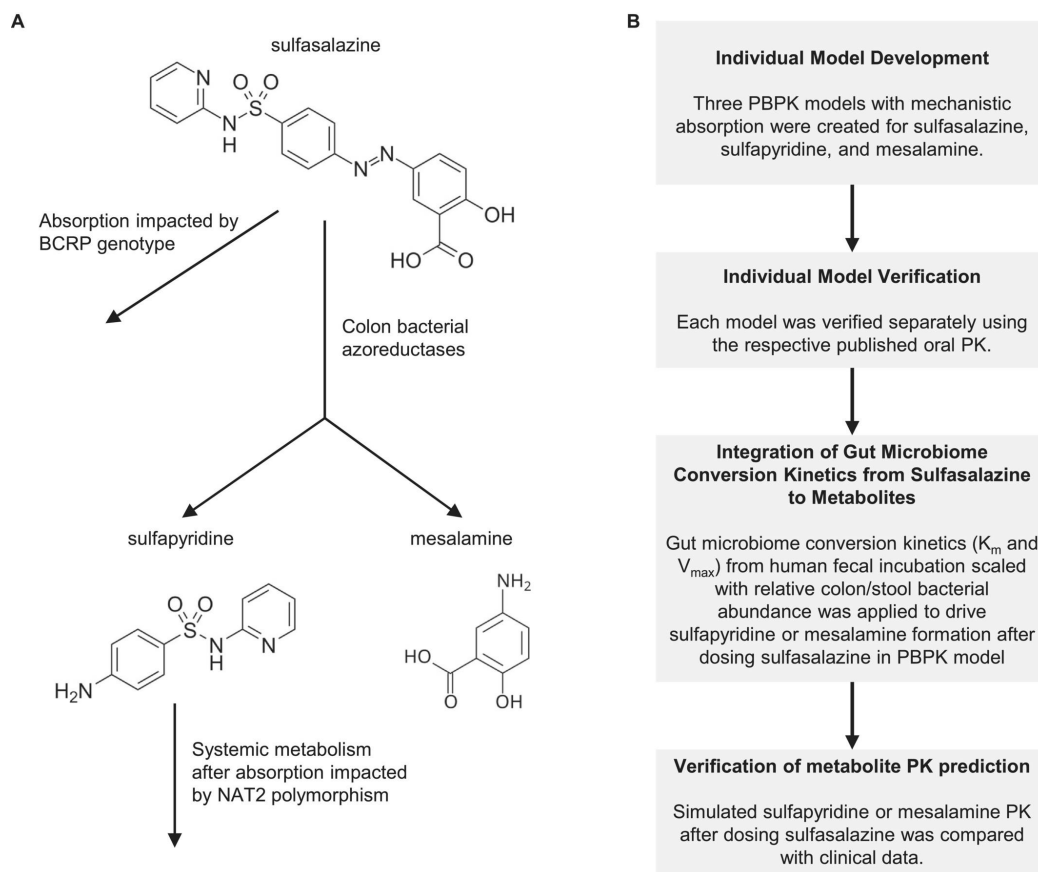


Figure 1: (A) Metabolism pathway of sulfasalazine and its metabolites relevant to this work. Breast cancer resist protein (BCRP), an efflux transporter with different geno

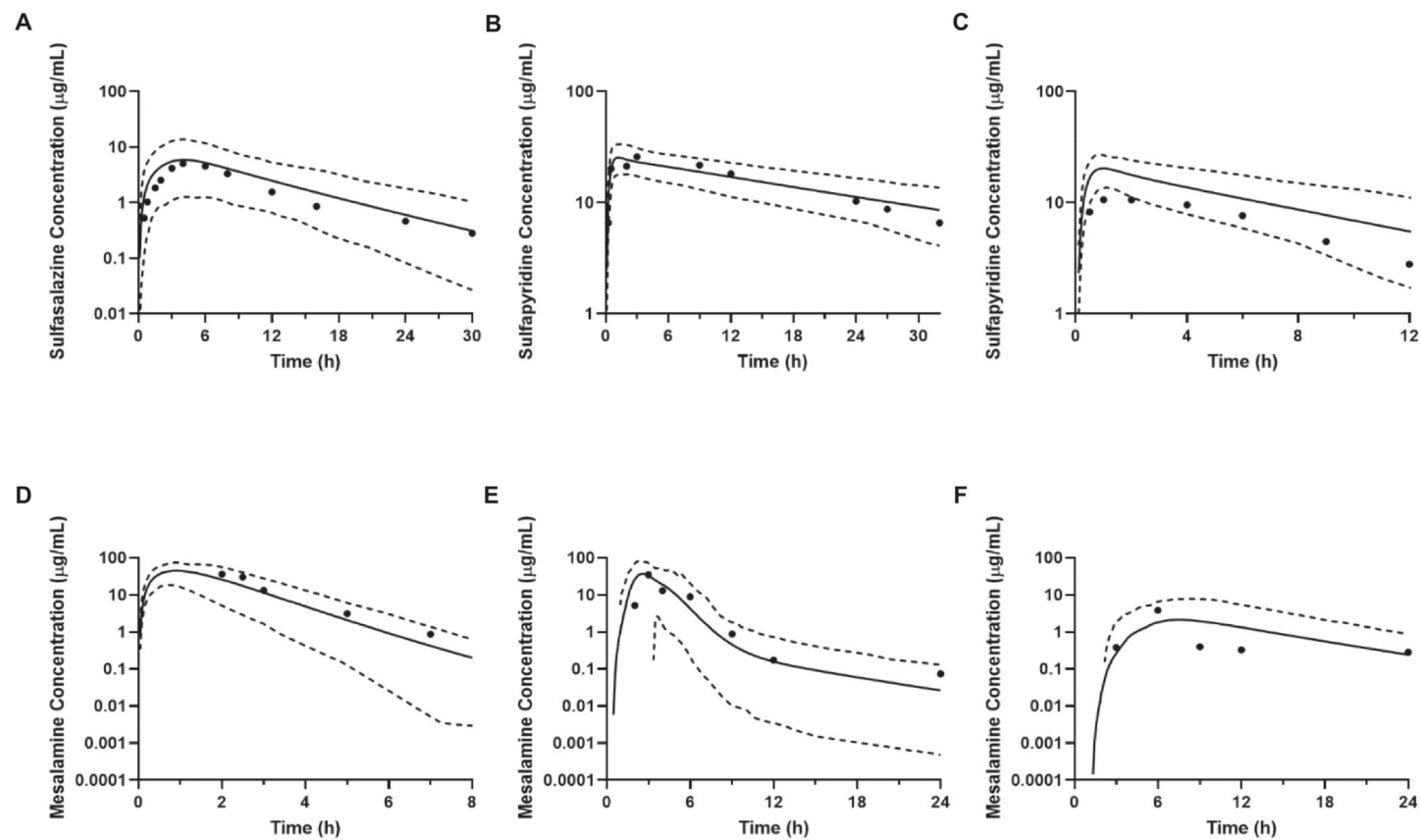


Figure 2: Representative simulated (solid black line) and observed (closed circles) concentration-time profiles of sulfasalazine (A), sulfapyridine (B, C), and mesalamine

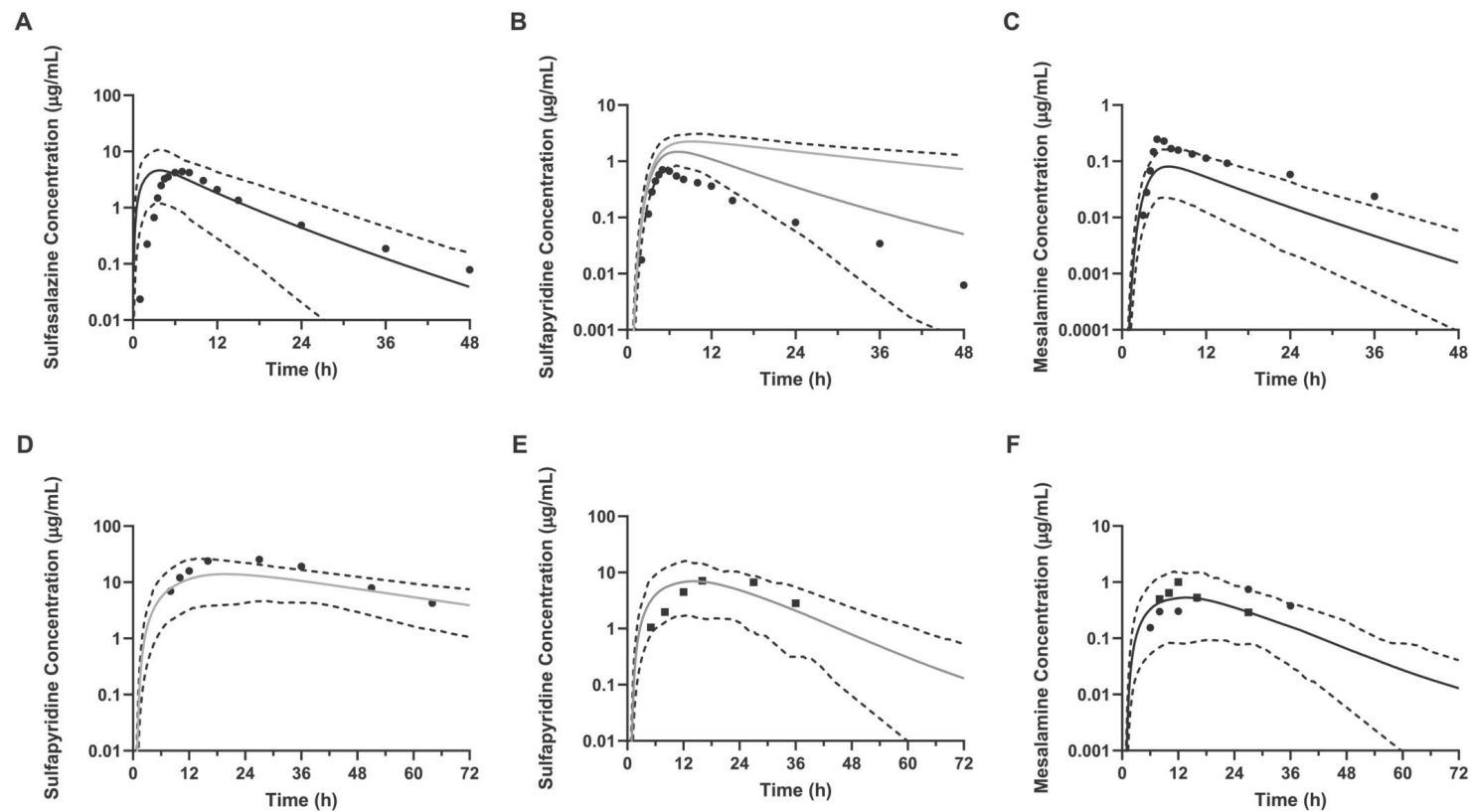


Figure 3: Representative simulated (solid black line) versus mean or individual observed concentration-time profiles of sulfasalazine and its metabolites after dosing heal

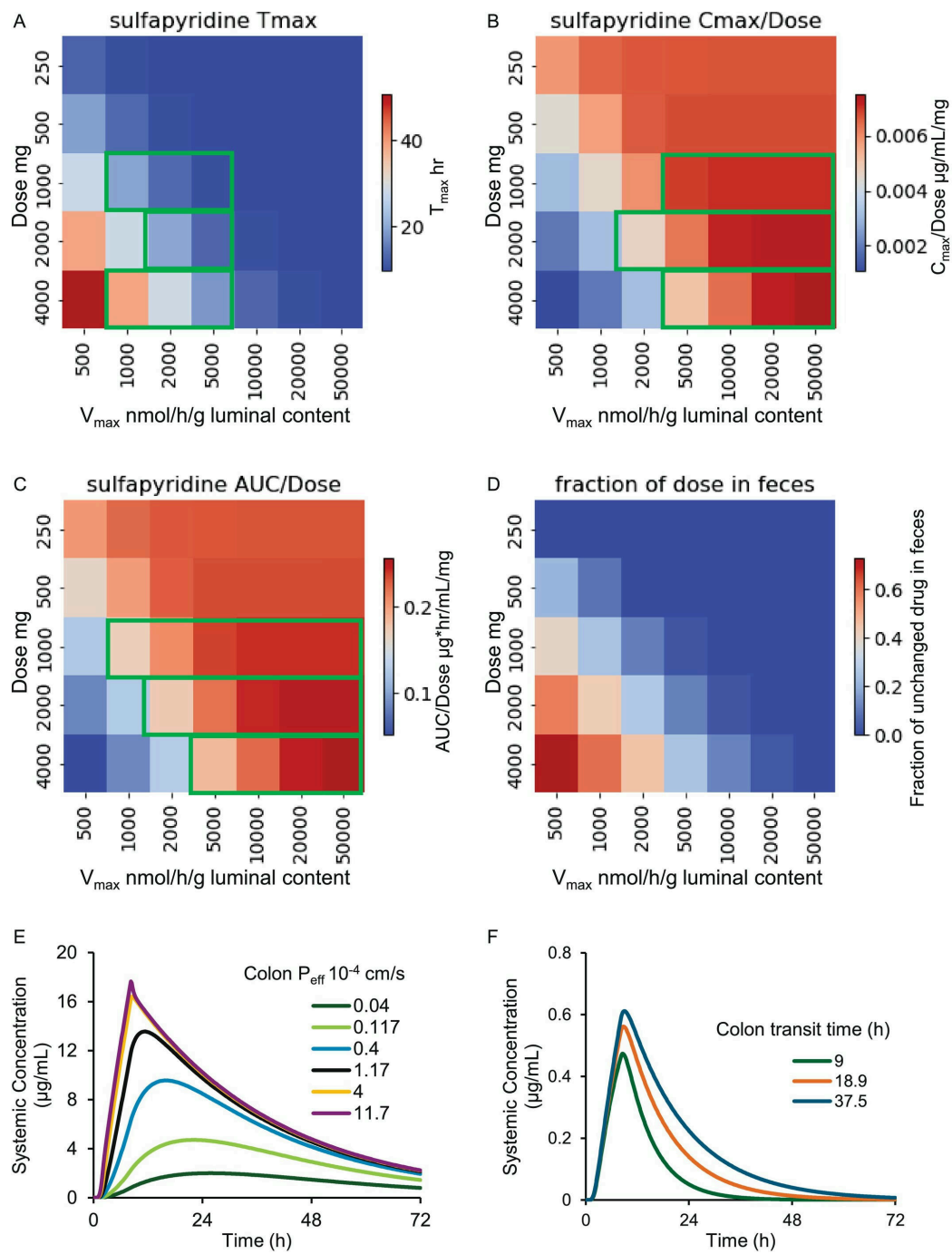


Figure 4: Sensitivity analysis of sulfapyridine or mesalamine PK after dosing sulfasalazine to different PBPK parameters. Sulfapyridine (A) T_{max} , (B) dose-normalized C_{max} , (